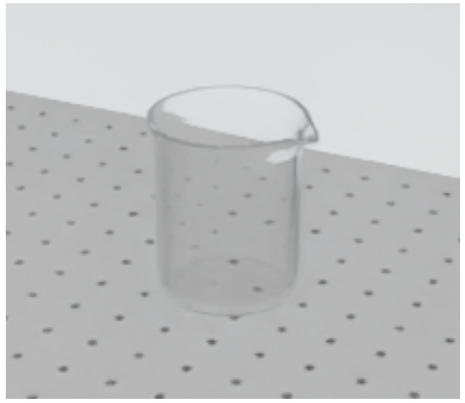
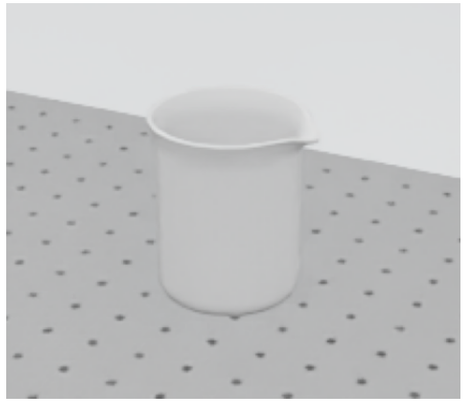
Transparent beaker perception fails with its observed depth

Official FoundationPose on a static MATTERIX fixture · no robot control · whole-material control, not opacity alone

Original glass · RGB



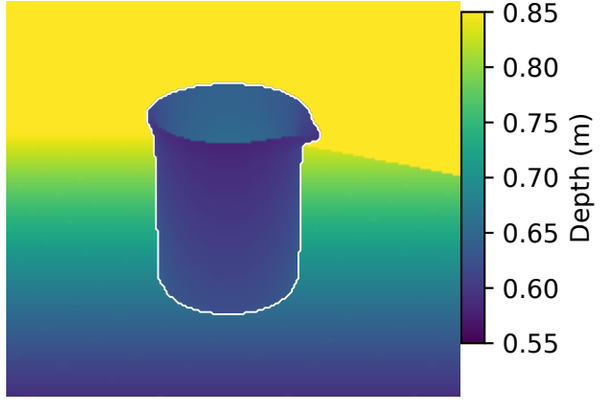
Opaque material control · RGB



Depth · median 0.775 m

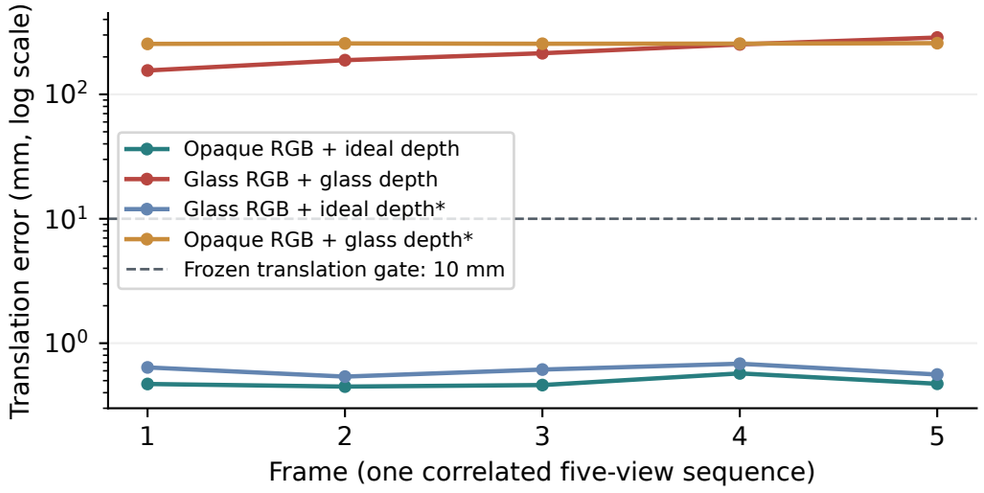


Depth · median 0.613 m



Same CAD, camera, weights and initial mask

Input channels	Pass	Translation error (mm)
Opaque RGB + ideal depth	5 / 5	0.45–0.57
Glass RGB + glass depth	0 / 5	155.50–286.14
Glass RGB + ideal depth*	5 / 5	0.54–0.68
Opaque RGB + glass depth*	0 / 5	254.40–257.86



* Post-hoc channel substitutions. Ideal depth and the initial oracle mask are privileged simulator inputs, not a deployed repair.

Pose acceptance also requires ADD-S ≤ 10% of sampled mesh diameter. Five correlated frames do not constitute five independent trials.